Corrected MCQs

1. **B.** The first vector is twice the second. Their scalar product is not available in any base; collinearity does not depend on a metric.
2. **A.** With the first two non-collinear vectors, the linear combination characterizes their plane The null vector is a particular case, not the only one.
3. **C.** Non-collinear directors exclude parallelism and confusion. The intersection remains to be established: the lines may also be non-coplanar.
4. **B.** A basis guarantees the existence and uniqueness of decomposition. the basics, which do not need to be orthonormized.
5. **C.** The interrupted trait retains the edge and indicates that it is hidden. Clearing the edge would remove information on the solid.
6. **B.** The directions AB, AC the direction of the plan. A relationship $AM = sAB + tAC$ therefore proves belonging.
7. **B.** Two planes of the same direction are parallel in the broad sense. distinguish planes combined and strictly parallel.
8. **A.** We average each pair of coordinates: $(1;2;3)$. This midpoint formula does not does not even require orthonormality.
9. **C.** The perspective is a projection that loses depth. An intersection drawn does not prove not an intersection of the space lines.
10. **A.** Three non-coplanar vectors form a basis. Orthogonality and norm 1 are not required.

True/false. False: the straight (AB) and (CG) a cube is not coplanar. True: $\vec{0} = 0\vec{u}$. False: a projection can create apparent intersections; a calculation or geometric proof is necessary.

Corrected applications and support

Application 1. $\overrightarrow{JK} = 2\vec{u} - \vec{v} - \frac{1}{2}\vec{w}$; $\overrightarrow{IK} = \frac{3}{2}\vec{u} - \frac{3}{2}\vec{v}$.

Application 2. The first two drawings place the right in a common plane; the third one represents opposite edges of an unflatted tetrahedron. In a cube, (AB) and (CG) are no coplanar: the only plan containing A, B, C is that of the face $(ABCD)$which does not contain G.

Implementation 3. (OA) and (BC) Add "coplanar". Two straight lines distinct from the same plane, not parallel, are stinging.

Implementation 4. The first coordination requires that a possible relationship between $\vec{v}$ and $\vec{u}$ either 0, impossible since $\vec{v} \neq 0$. The combination is $(\lambda; \lambda + \mu; \mu) : \lambda = 2$, $\mu = 3$, so $m = 5$.

Application 5. It takes an orthonormal coordinate system. With two unit vectors of angle 60 °, the square of the norm of their sum is $1 + 1 + 2\cos 60° = 3$; the norm is $\overline{3}$. In the first passage, you can also see this length as a diagonal of diamond.

Implementation 6. $\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{AB}$ and $\overrightarrow{IK} = \frac{1}{2}\overrightarrow{AC}$: two non-collinear directions of the planes are parallel, So the planes are parallel. $O \in (IJK)$, then $\overrightarrow{IO} = \lambda\overrightarrow{IJ} + \mu\overrightarrow{IK}$. The coefficients $\overrightarrow{OB}$ and $\overrightarrow{OC}$ require $\lambda = \mu = 0$, while that of $\overrightarrow{OA}$ would be $-\frac{1}{2} = 0$, contradiction.

Space research

Manipulate four non-coplanar points, then draw. In the correction, distinguish this to provide assistance, to provide
three Chasles decompositions; to deepen, replace the midpoints with the same ratio $t \in]0; 1[$ and find parallelism.

Conduct and correction

Benchmarks for the teacher

The DS1 precedes S12 and S15: propose vector geometry in a given base, without require new coordinate techniques. Use a material cube, then a perspective cavalier; mark the hidden edges. Two non-parallel lines of space are not The norm by coordinates requires an orthonormal coordinate system. formalized parametric representations and plan equations will be processed later.

Corrected activity

1. By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = -\frac{1}{2}\vec{\imath} + \frac{1}{2}\vec{\jmath}$ and $\overrightarrow{IK} = -\frac{1}{2}\vec{\imath} + \frac{1}{2}\vec{k}$.

2. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would be $\frac{1}{2} = 0$, impossible in one base. These two vectors are therefore non-collinear; I, J, K define a plan.

3. The center of the cube checks $\overrightarrow{AO} = \frac{1}{2}(\vec{\imath} + \vec{\jmath} + \vec{k})$, therefore $\overrightarrow{IO} = \frac{1}{2}\vec{\jmath} + \frac{1}{2}\vec{k}$. If $O \in (IJK)$, there is s, t such that $\overrightarrow{IO} = s\overrightarrow{IJ} + t\overrightarrow{IK}$. By comparing the base coefficients,

$$0 = -\frac{s+t}{2}, \qquad \frac{1}{2} = \frac{s}{2}, \qquad \frac{1}{2} = \frac{t}{2}.$$

The last two require $s = t = 1$, while the first would ask $s + t = 0$. Contradiction : $O \notin (IJK)$.

4. The cavalier perspective is a projection on a sheet: separate points may have the It helps to conjecture, but an apparent superimposition does not prove belonging to a plan.

E1. The three coordinates of $\vec{v}$ are worth three times that of $\vec{u}$: $\vec{v} = 3\vec{u}$. The two vectors are therefore Collinear.

E2. The midpoint has its coordinates $(\frac{0+4}{2}, \frac{1-1}{2}, \frac{2+0}{2}) = (2, 0, 1)$. The difference in coordinates gives $\overrightarrow{AB} = (4, -2, -2)$; in the orthonormal coordinate system, $AB = \sqrt{16+4+4} = 2\sqrt{6}$.

E3. We calculate $2\vec{u} + 3\vec{v} = (2, 0, 2) + (0, 3, 3) = (2, 3, 5) = \vec{w}$. Thus $\vec{w}$ belongs to the vector plan led by the first two vectors: the three are coplanar.

E4. The first two vectors are non-colonial. If the third was their combination $\lambda(1, 0, 1) + \mu(0, 1, 1)$, the first two coordinates would require $\lambda = \mu = 1$, but the third would be worth $2 \neq 0$. The three vectors are non-coplanar and thus form a base of space.

E5. $\overrightarrow{AB} = (2, -1, 2)$ and $\overrightarrow{AC} = (4, -2, 4) = 2\overrightarrow{AB}$. Like $A \neq B$, right (AB) is defined and this collinearity proves $C \in (AB)$.

Corrected from DM10

1. By Chasles and the definitions of midpoints,

$$\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{AC} - \frac{1}{2}\overrightarrow{AB} = \frac{1}{2}\overrightarrow{BC}, \qquad \overrightarrow{IK} = \frac{1}{2}\overrightarrow{AD} - \frac{1}{2}\overrightarrow{AB} = \frac{1}{2}\overrightarrow{BD}.$$

2. Points B, C, D are not aligned because the tetrahedron is not flattened. $\overrightarrow{BC}, \overrightarrow{BD}$ are therefore non-colonial, as well as $\overrightarrow{IJ}, \overrightarrow{IK}$. Plans have the same direction. They are distinct: if $I \in (BCD)$, right $(BI) = (BA)$ would be included in this plan, which would place A, contradiction. They are therefore strictly parallel.

3. In the base $(\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD})$, we have $\overrightarrow{IG} = (-\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, $\overrightarrow{IJ} = (-\frac{1}{2}, \frac{1}{2}, 0)$ and $\overrightarrow{IK} = (-\frac{1}{2}, 0, \frac{1}{2})$. A relationship $\overrightarrow{IG} = s\overrightarrow{IJ} + t\overrightarrow{IK}$ imposes $s = t = \frac{1}{2}$ by the last two coordinates; the first would then be $-\frac{1}{2}$, and not $-\frac{1}{4}$. So $G \notin (IJK)$.

4. In this particular case, $I(2,0,0)$, $J(0,2,0)$ and $K(0,0,2)$. We find $IJ = \sqrt{(-2)^2 + 2^2} = 2\sqrt{2}$and of Same $IK = JK = 2$ $\overline{2}$. The triangle is equilateral. This metric conclusion does not apply to any non-flatted tetrahedron: it uses the coordinates of the particular case.

Corrected work in the manual or an anal

In this exercise, I is the midpoint of $[AB]$ and J that of $[CD]$.

1. Plans (AIE) and (BIG) both contain (AB). They are distinct: G does not belong to the Plan of the face $ABFE$. Their intersection is therefore exactly the right (AB).

2. Points A, D, I are three non-aligned points on the face $ABCD$; the same applies to B, J, C. The two plans are therefore confused with (ABC). The intersection is the integer plane, not a straight one.

3. The plan (HEF) is that of the upper face; (BJC) is that of the lower face. Strictly parallel: their intersection is empty.